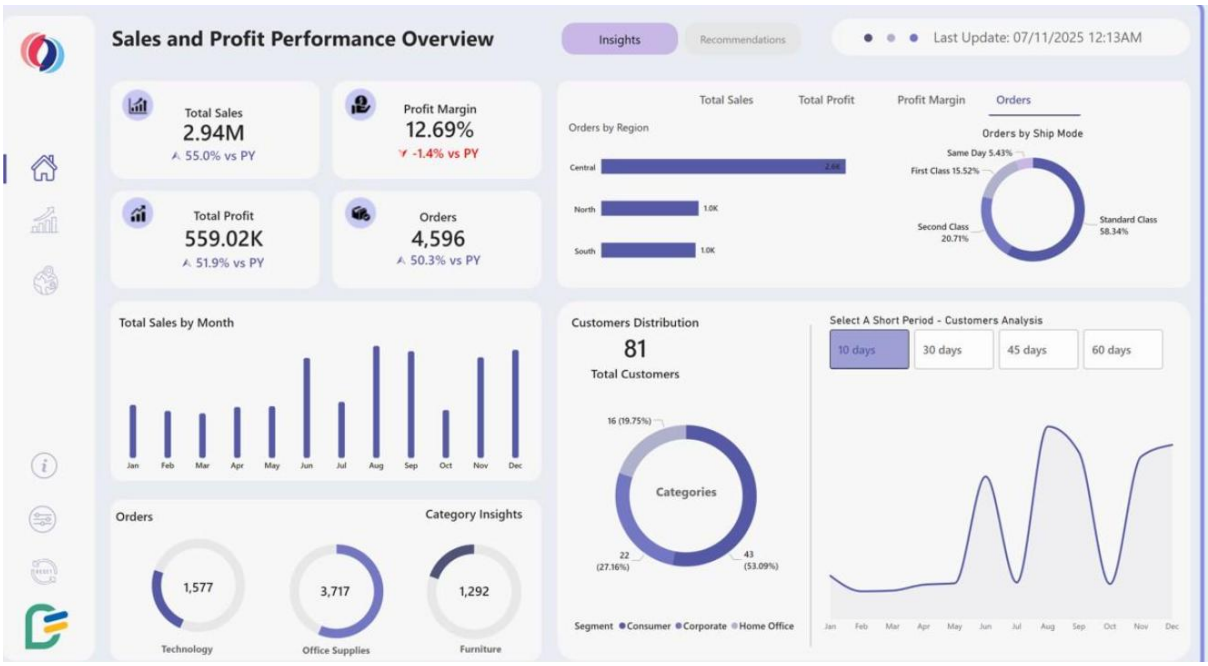
Practical 12

Problem definition:

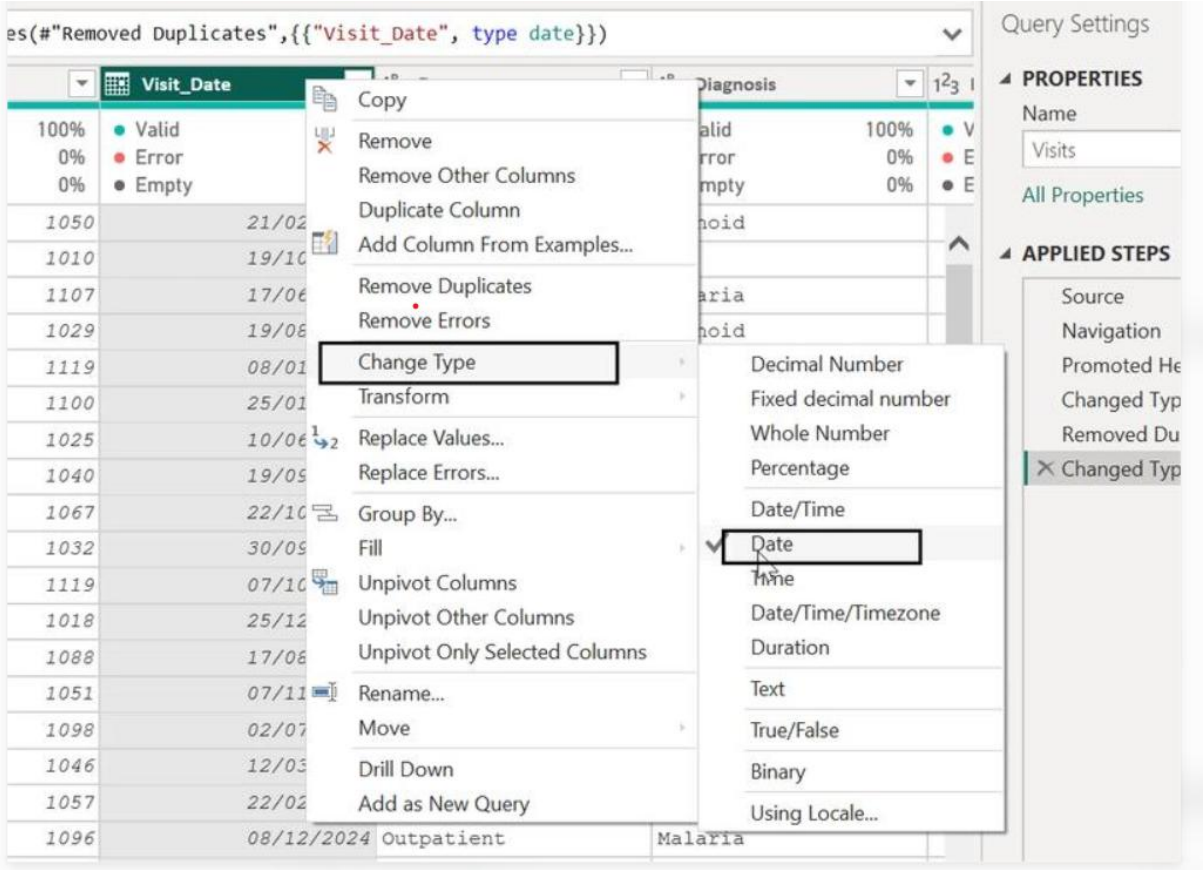
You are working as a Data Visualization Analyst at a retail analytics company. Management wants to monitor sales performance across products, regions, and customer segments through an interactive dashboard. Static reports are insufficient for decision-making. Your task is to design an interactive dashboard that allows management to explore sales data and identify key business trends.

Observation / Output:

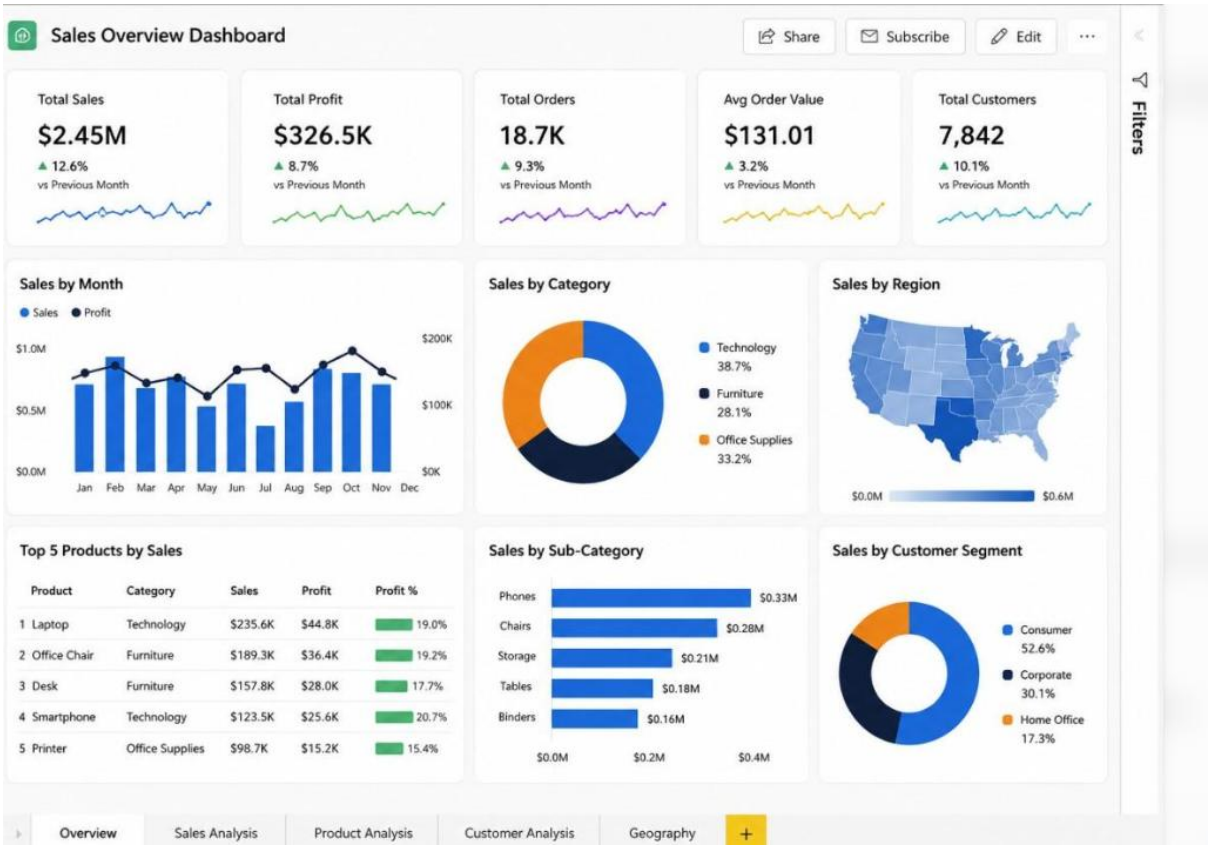
Task 1 : Import sales dataset into Tableau / Power BI.



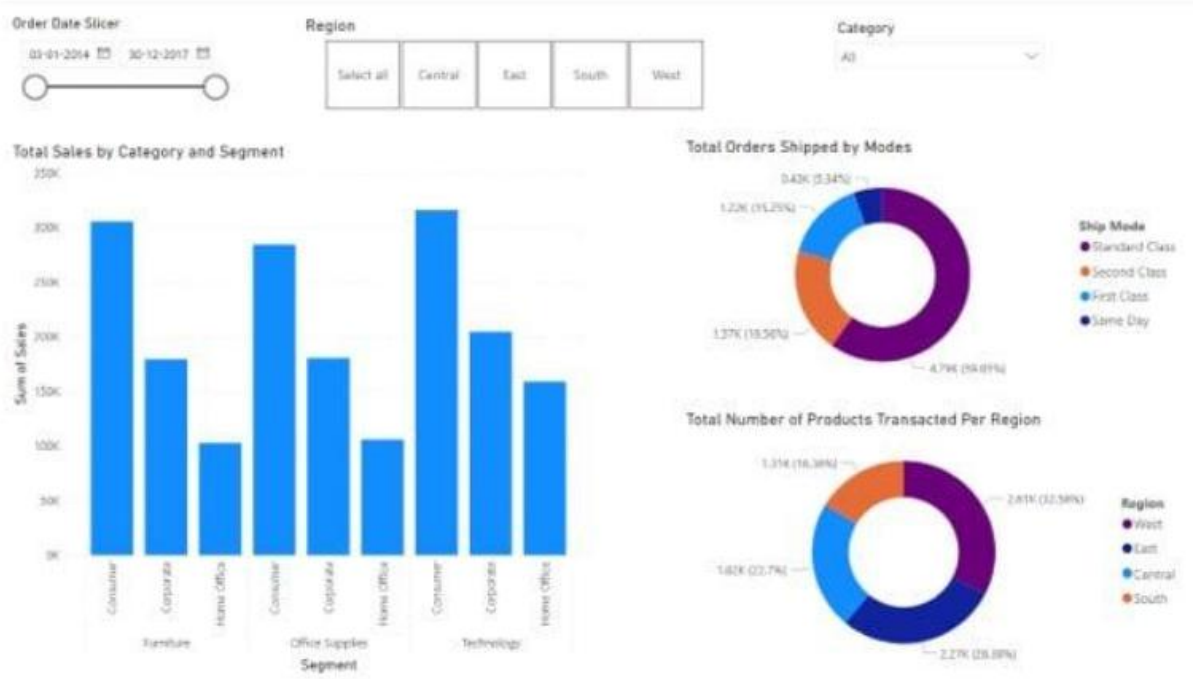
Task 2: Clean and validate the dataset.



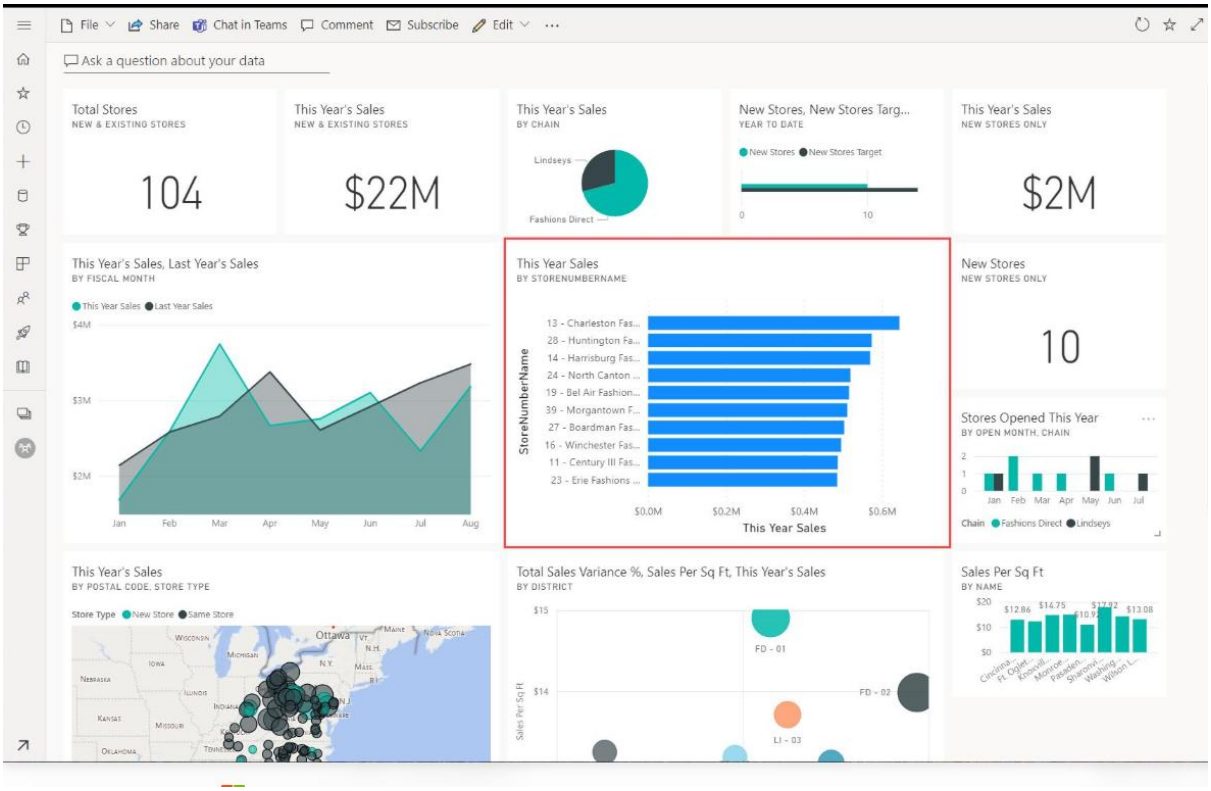
Create KPIs for Total Sales, Profit, and Orders.



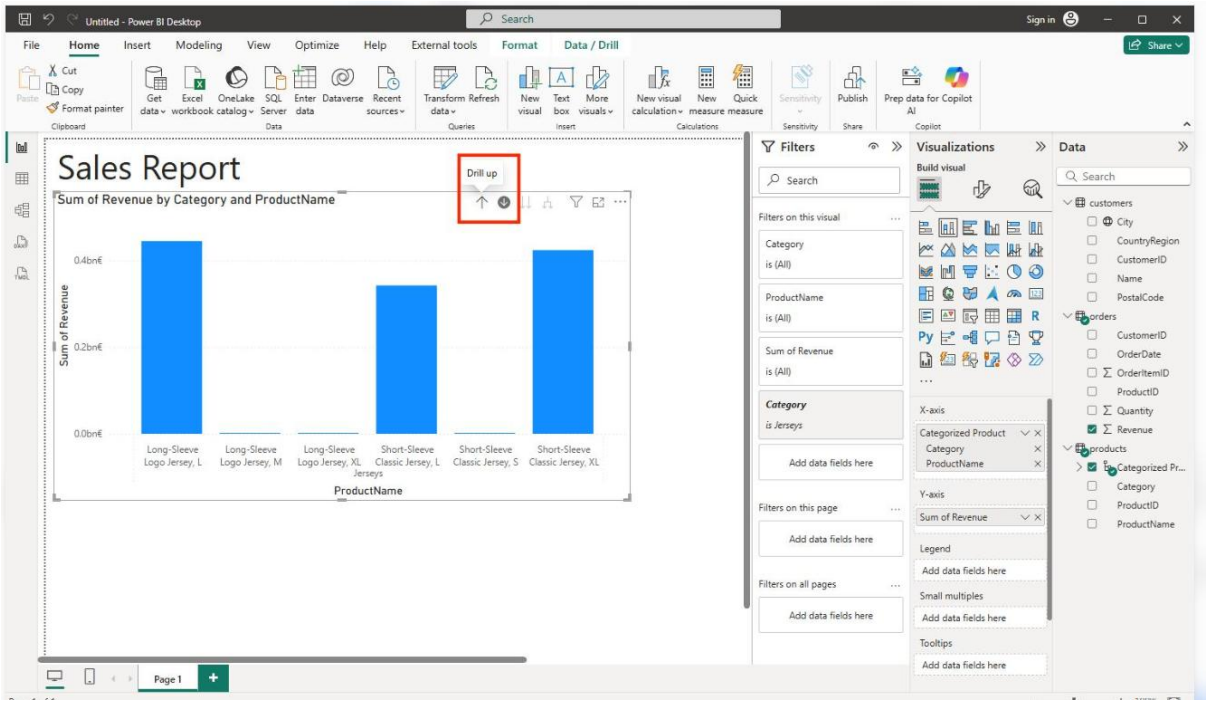
Design filters for Region, Category, and Time.



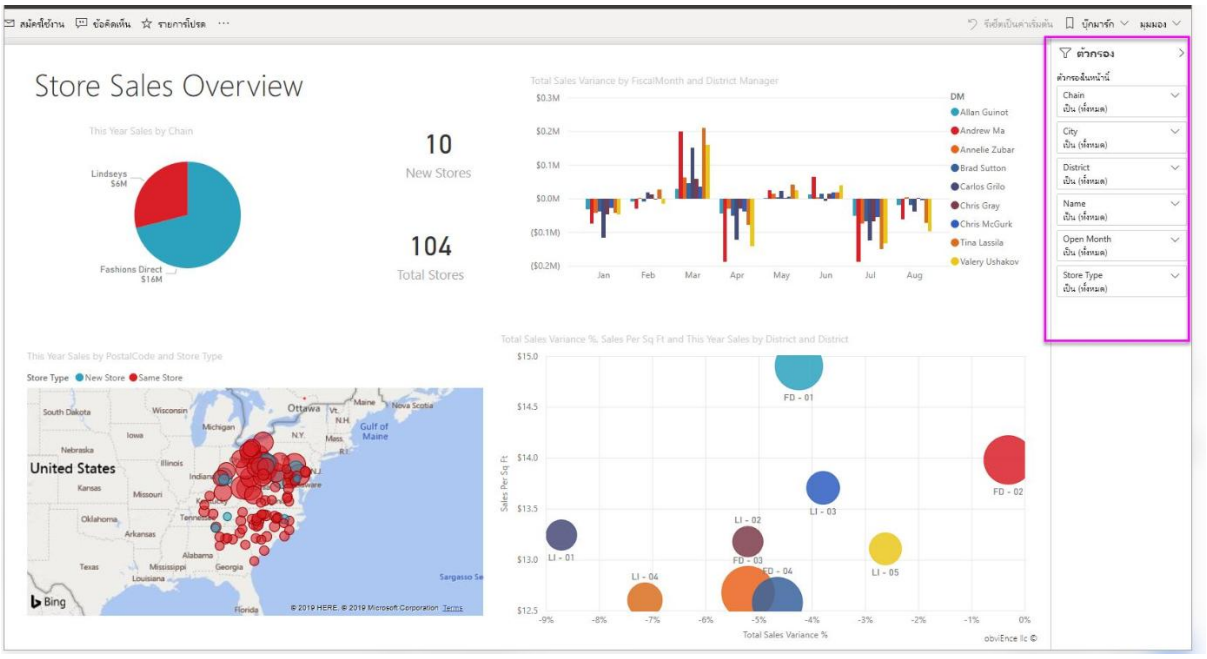
Create interactive charts for sales analysis.



Develop drill-down functionality.



Add dashboard interactions.



Create summary cards. Publish dashboard. Interpret business insights.



Key Questions /Analysis:

Q1. What is the role of interactive dashboards in business analytics?

Interactive dashboards help businesses analyze and monitor sales performance in real time. They present data through charts, graphs, and KPIs for easy understanding. Users can apply filters and explore trends to identify problems and opportunities. This supports faster, data-driven decision-making.

Q2. How do filters improve visual exploration?

Filters allow users to view specific data based on region, category, date, or customer segment. They help users focus on relevant information and compare different data groups. Filters make data exploration faster and easier. This helps identify trends and patterns for better decision-making.

Q3. What are KPI indicators?

KPI (Key Performance Indicators) are measurable values used to evaluate business performance. They help track progress toward organizational goals. Examples include Total Sales, Profit, Number of Orders, and Customer Retention Rate. KPIs help management monitor performance and make informed decisions.

Q4. Why are dashboards preferred over static reports?

Dashboards are preferred over static reports because they provide interactive and dynamic data visualization. Users can apply filters, explore trends, and view updated information easily. Static reports display fixed information and offer limited exploration. Dashboards support faster analysis and better decision-making.

Q5. How does user-driven exploration support decision-making?

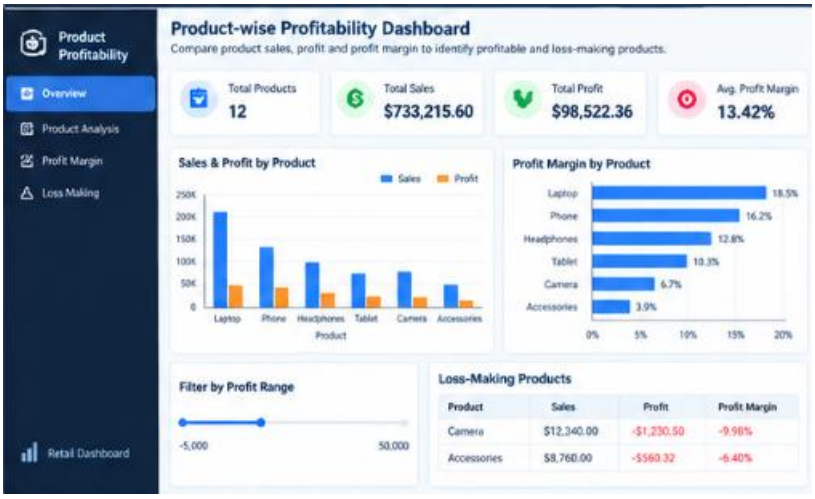
User-driven exploration allows users to interact with data using filters, charts, and drill-downs. It helps them identify trends, patterns, and business problems. Users can compare different regions, products, and time periods. This supports informed decisions and improves business performance.

Supplementary Problems:

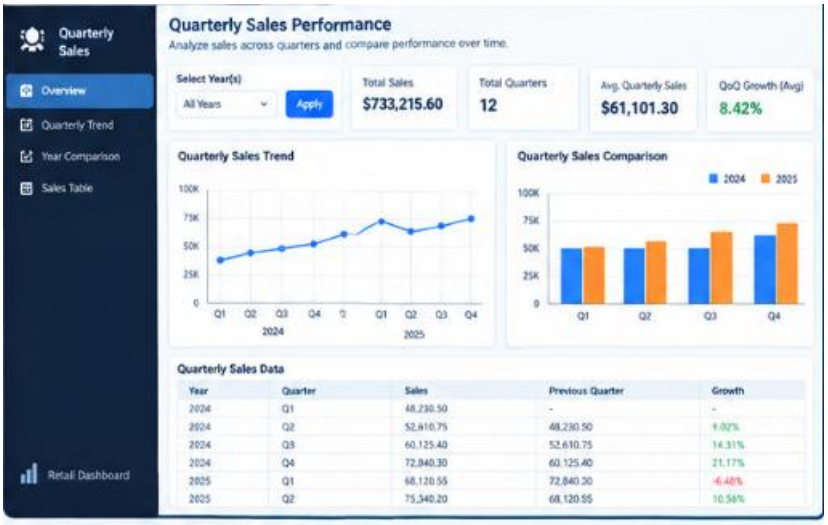
Add customer segmentation analysis.



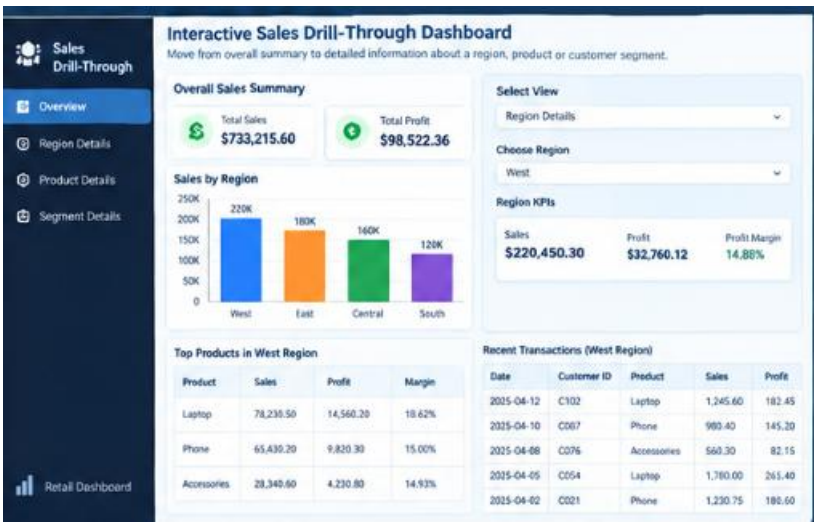
Create product-wise profitability dashboard.



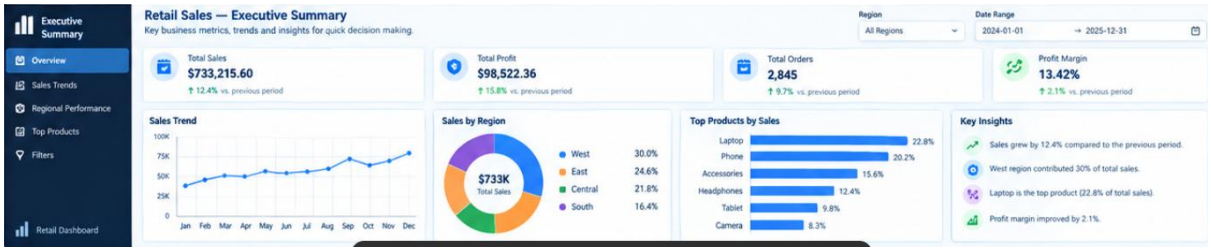
Compare quarterly sales performance.



Implement drill-through pages.



Create executive summary dashboard.



Tools / Technology used:

Tableau / Power BI | Excel / CSV Dataset | Python

Conclusion:

In this practical, we learned how to create an interactive customer sales dashboard using data visualization tools. We analyzed sales, profit, and orders using KPI indicators. We used filters to explore data by region, category, and time. Interactive charts and drill-down features helped us identify sales trends and patterns. We understood the importance of dashboards in business analytics and decision-making. We also learned how customer segmentation and profitability analysis can provide useful business insights. Overall, this practical improved our understanding of interactive analytics and business intelligence reporting.

Github Repo:

https://github.com/tushar-2606/BDA-SHP-2026